

Design and Verification of a Pipelined RV32I RISC-V Processor with Dynamic Branch Prediction on FPGA

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PROJECT OBJECTIVE

- Verify and implement the processor on FPGA
- Integrate dynamic branch prediction
- Implement a 5-stage pipeline with hazard handling
- Design a 32-bit RV32I RISC-V processor

INTRODUCTION

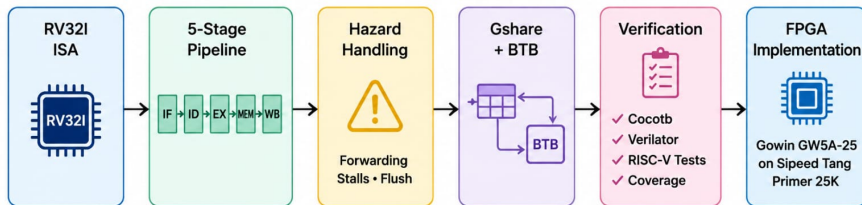
- RISC-V is an open and flexible ISA
- Pipelining improves processor performance
- Pipeline hazards affect execution efficiency
- Dynamic branch prediction reduces branch penalties
- FPGA enables practical hardware implementation

LITERATURE SURVEY

Sl No.	Title	Author/Journal/Year	Methodology	Remarks
1	Design and Implementation of 5-Stage Pipelined RISC-V Processor on FPGA	IEEE Conference Publication, 2024	5-stage RISC-V pipeline on FPGA	Demonstrates practical FPGA implementation of a pipelined RISC-V processor
2	A Study of Branch Prediction Strategies	J.E.Smith, IEEE/ISCA, 1981	Branch prediction techniques	Establishes fundamentals of branch prediction
3	Design and Analysis of a Multi Clocked Pipelined Processor Based on RISC-V	IEEE Conference Publication, 2022	Pipelined RISC-V architecture	Studies pipelined RISC-V processor implementation
4	Understanding Branches and Designing Branch Predictors for High-Performance Microprocessors	IEEE, 1996	Branch predictor design	Discusses branch behavior and predictor design considerations

5	An Efficient Implementation of a TAGE Branch Predictor for Soft Processors on FPGA	IEEE Conference, 2019	FPGA branch prediction	Shows the importance of efficient branch prediction in FPGA soft processors
6	Design and Implementation of a Pipelined RV32IMC Processor with Interrupt Support	IEEE Conference Publication, 2020	Pipelined RV32 processor	Demonstrates an extended RISC-V pipeline implementation
7	Two-Level Adaptive Training Branch Prediction	Yeh Patt, MICRO, 1991	Two-level dynamic prediction	Uses run-time branch history for improved prediction

METHODOLOGY



SCHEDULE OF THE WORK

- Literature survey and architecture design
- RTL implementation
- Pipeline and hazard handling
- Integration of branch predictors
- Verification and testing
- FPGA implementation
- Performance analysis

REFERENCES

- RISC-V International, RISC-V Instruction Set Manual
- Patterson and Hennessy, Computer Organization and Design
- Hennessy and Patterson, Computer Architecture: A Quantitative Approach
- J. E. Smith, A Study of Branch Prediction Strategies
- Verilator Documentation
- Cocotb Documentation

THANK YOU